

# DengueTriage-ES: An Expert System for Dengue Severity Triage

Name: — *to be completed* —

Index Number: — *to be completed* —

Source Code: *DengueTriage-ES-code.zip (submitted with this report)*

## 1. Introduction

About half of the world's population is now at risk of dengue, and 2024 set a record with 14.6 million reported cases and more than 12,000 deaths [1]. Sri Lanka is among the hardest-hit countries; its largest outbreak, in 2017, produced more than 186,000 reported cases [2].

The difficulty with dengue is timing. The fever lasts two to seven days, and the danger comes after it: around defervescence, usually on days three to seven, capillaries start to leak plasma for 24 to 48 hours, which the guidelines call the critical phase [3]. A patient who feels better as the fever settles can be on the way into shock. In an epidemic, doctors who are not dengue specialists have to decide again and again whether a patient can go home, needs admission, or needs emergency treatment.

DengueTriage-ES makes that decision in the way the WHO guideline sets out. From a patient's clinical findings, the system:

- checks whether the dengue case definition is met,
- classifies the case as dengue without warning signs, dengue with warning signs, or severe dengue,
- assigns WHO management group A (home care), B (in-hospital care) or C (emergency treatment),
- states the guideline's action plan, detects the start of the critical phase and checks the discharge criteria, and
- explains every conclusion by listing each rule used and the guideline page it came from.

The system is limited to adult patients and assesses one point in time. A finding that was not assessed is treated as absent, and the system has not been clinically validated. It was built for teaching and must not be used to treat patients.

## 2. System Design

### 2.1 Knowledge engineering and sources

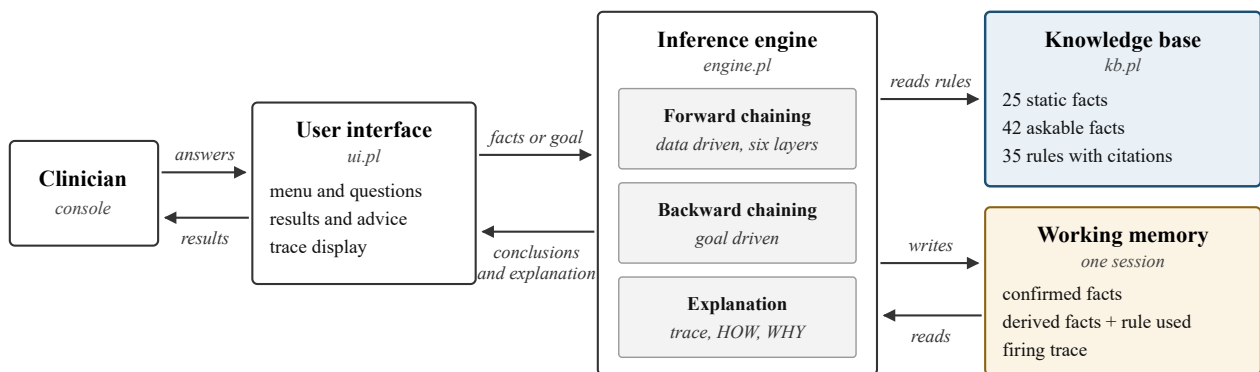
No domain expert was interviewed; as the brief allows, the knowledge was taken from published clinical guidelines instead. All 35 rules come from the WHO guideline *Dengue: Guidelines for Diagnosis, Treatment, Prevention and Control* [3], which defines both the case classification and the A/B/C management groups. The Sri Lanka national guideline for adults [4] was read for the local setting; because it uses the older dengue fever / dengue haemorrhagic fever (DF/DHF) terms, the rules follow the WHO classification directly. CDC guidance [5] was used to cross-check wording and thresholds. The process followed is summarised below.

| Stage                 | What was done                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------|
| Domain selection      | Chose dengue triage: a frequent, high-stakes decision whose criteria are already written down |
| Domain study          | Studied the WHO 2009 guideline, the Sri Lanka adult guideline and the CDC clinical pages      |
| Knowledge acquisition | Extracted every decision criterion word for word from WHO Figure 1.4 and Chapter 2            |

| Stage                    | What was done                                                                       |
|--------------------------|-------------------------------------------------------------------------------------|
| Knowledge representation | Production rules in SWI-Prolog, each stored next to a source/3 fact citing its page |
| Facts and rules          | 67 facts and 35 rules, arranged in six layers                                       |
| Inference                | Layered forward chaining and goal-driven backward chaining over the same rules      |
| Testing                  | 32 automated tests; each defect found was fixed in the rule base                    |
| Explanation              | Cited rule trace, HOW proof trees and WHY answers during questioning                |

Guideline text was turned into rules using four fixed conventions. A list of alternatives became a class of facts tested with `any_of`. “Two of the following” became `atleast(2, Class)`. A numeric threshold became a yes/no fact that the clinician answers, which keeps arithmetic out of the rule base entirely. An exclusion such as “without warning signs” became an explicit `absent(...)` premise. For example, the WHO text states that a patient “is considered to have shock if the pulse pressure ... is  $\leq$  20 mm Hg ... or he/she has signs of poor capillary perfusion” [3, p. 28]. That sentence became two rules: **R03** IF pulse pressure  $\leq$  20 mmHg THEN shock, and **R04** IF signs of poor capillary perfusion THEN shock.

## 2.2 Architecture



**Figure 1:** System architecture

Everything the clinician sees passes through the console interface, which hands the answers, or a chosen goal, to the inference engine. The engine takes its rules from the knowledge base and writes every confirmed and derived fact to working memory, together with the rule that produced it. The explanation facility builds its traces and proof trees from those stored justifications. The engine itself contains no clinical knowledge, so a different knowledge base could be loaded without changing it.

| Component        | File                                                 | Role                                                    |
|------------------|------------------------------------------------------|---------------------------------------------------------|
| User interface   | <code>ui.pl</code>                                   | Menu, questions, results, advice and trace display      |
| Inference engine | <code>engine.pl</code>                               | Working memory, forward and backward chaining           |
| Explanation      | <code>engine.pl</code>                               | Rule trace, HOW proof trees, WHY goal chains            |
| Knowledge base   | <code>kb.pl</code>                                   | 67 facts; 35 rules, each with its citation; advice text |
| Tests            | <code>scenarios.pl</code> ,<br><code>tests.pl</code> | 12 patient scenarios and 32 automated tests             |

### 3. Knowledge Base

#### 3.1 Facts

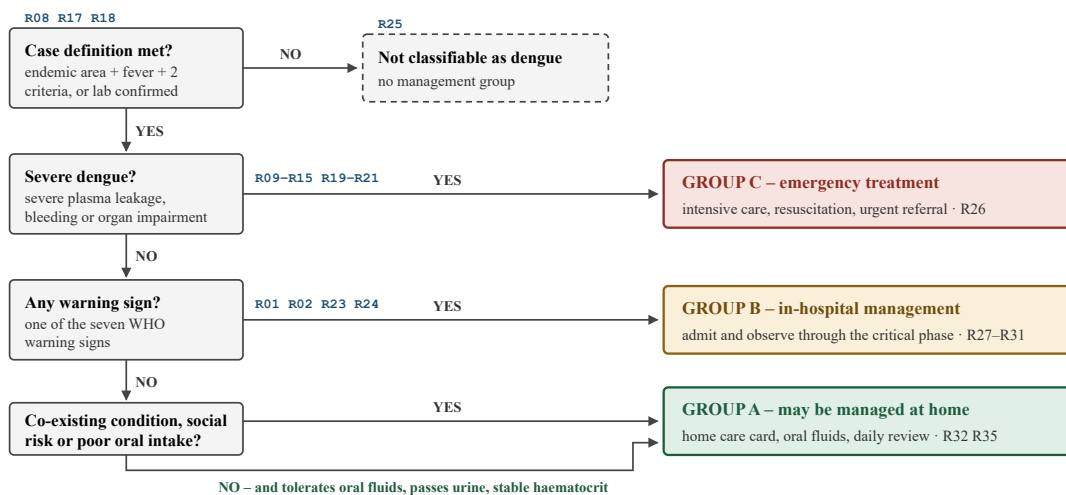
The knowledge base holds 67 facts. The 42 **askable facts** are the questions the system can put to the user; each stores its question text and the interview section it belongs to, for example `askable(narrow_pulse_pressure, 'Is the pulse pressure 20 mmHg or less?', severe)`. The 25 **static facts** define the vocabulary of the domain and are grouped into five classes, so that one rule can refer to a whole class:

| Class                             | Members                                                                                                                          |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| probable-dengue criteria          | nausea_vomiting, rash, aches_pains, tourniquet_positive, leukopenia, warning_sign_present                                        |
| WHO warning signs                 | abdominal_pain, persistent_vomiting, fluid_accumulation, mucosal_bleeding, lethargy_restlessness, liver_enlargement, lab_warning |
| co-existing conditions            | pregnancy, infancy, old_age, obesity, diabetes_mellitus, renal_failure, chronic_haemolytic_disease                               |
| social circumstances              | lives_alone, lives_far_from_facility                                                                                             |
| signs of severe organ involvement | ast_alt_ge_1000, impaired_consciousness, heart_impairment                                                                        |

#### 3.2 Rules

In the code, a rule is the term `rule(Id, Layer, Conclusion, Premises)`, written directly above a `source(Id, Document, Page)` fact that cites it. A premise can be a fact, `absent(Fact)`, `any_of(List)` or `atleast(N, Class)`.

Figure 2 shows the decision the rules encode. The order of the checks is the guideline's: severity comes before warning signs, and warning signs before host factors, so the most urgent group always applies. The guideline does not state in words that Group C overrides Group B, but its ordered triage implies it, so rules R28–R31 each include “NOT severe dengue”. This is the only design choice in the rule base that is not a direct transcription.



**Figure 2:** The clinical decision encoded by the rule base, with the rules used at each step

The 35 rules fall into six groups, one for each layer used by the inference engine (Section 4.1). The full list is in Annex A.

| Group (layer)                              | Rules   | Decides                                                                                       | Source [3]                   |
|--------------------------------------------|---------|-----------------------------------------------------------------------------------------------|------------------------------|
| A. Composite findings (1)                  | R01–R07 | Warning sign present; shock; co-existing or social risk; laboratory confirmation              | Fig. 1.4; s.2.1.4; s.2.3.2.2 |
| B. Case definition and severe criteria (2) | R08–R16 | Probable dengue; severe plasma leakage, bleeding or organ impairment; start of critical phase | Fig. 1.4; s.2.1.2; s.2.3.3   |
| C. Dengue case (3)                         | R17–R18 | Whether the case definition is met                                                            | Fig. 1.4                     |
| D. Severity (4)                            | R19–R21 | Severe dengue                                                                                 | s.2.1.4                      |
| E. Classification (5)                      | R22–R25 | The WHO case class                                                                            | Fig. 1.4                     |
| F. Management and advice (6)               | R26–R35 | Group A, B or C; discharge; drugs to avoid; daily review                                      | s.2.3.2.1–2.3.2.3; Textbox F |

## 4. Inference Engine and Explanation

### 4.1 Forward chaining (full consultation)

Forward chaining begins with every answer the user gave. It repeatedly looks for a rule whose premises are all satisfied and whose conclusion is not yet known, fires it, and stops when a full pass fires nothing new. The rules are divided into the six layers of Section 3.2, and each layer is completed before the next begins. This is needed because of negation: without layers, R24 (“dengue without warning signs”) could fire before R02 had found a warning sign, and the answer would depend on the order the rules appear in the file. Since every `absent (...)` premise refers only to a lower layer, the result does not depend on rule order.

Scenario T04 (dengue shock syndrome) shows the layers at work. Layer 1 finds a warning sign and shock (R02, R03); layer 2 establishes probable dengue and severe plasma leakage (R08, R09); layer 3 confirms the dengue case (R17); layer 4 concludes severe dengue (R19); layer 5 classifies it (R22); and layer 6 assigns Group C and adds the NSAID warning (R26, R34).

### 4.2 Backward chaining (testing one hypothesis)

Here the starting point is a hypothesis chosen from a menu of five, such as “Does this patient have severe dengue?”. The engine looks up every rule whose conclusion matches the hypothesis and tries to prove that rule’s premises in turn, putting a question to the user only when a proof reaches a fact nobody has supplied yet. A counting premise stops as soon as enough members are found.

```

Goal: severe dengue          R19: dengue case AND severe plasma leakage
├─ dengue case              R17 <- R08: endemic area, fever, 2 criteria
│   ask endemic area -> yes; ask fever -> yes
│   ask nausea/vomiting -> yes; ask rash -> yes    (2 found; 4 not asked)
└─ severe plasma leakage    R09 <- shock <- R03
    ask pulse pressure <= 20 mmHg -> yes
Result: severe dengue PROVED after 5 questions (forward mode asks 42)

```

### 4.3 Explanation

Three facilities share the justifications stored in working memory. At the end of every consultation the system prints a numbered trace of every fired rule in sequence, with its premises (including those satisfied because something was absent), its conclusion, and its guideline section and page. Menu

option 5 prints a HOW proof tree for any single conclusion. During backward chaining, typing **w** at a question shows WHY it is being asked: the chain of goals the engine is trying to prove at that moment. If a goal cannot be proved, the system says that no rule's conditions were met and that the condition has not been medically ruled out.

## 5. User Interface

DengueTriage-ES is operated from a console window. A text interface was chosen because the main output, the reasoning trace, is long and indented, and it reads better as text. The main menu (Figure 3) offers both consultation modes, 12 built-in scenarios that run without any typing, the trace, the HOW explanation, a browser for the knowledge base, the test suite and help.

```
MAIN MENU
-----
1 New consultation (forward chaining, data driven)
2 New consultation (backward chaining, goal driven)
3 Run a built-in scenario
4 Show the reasoning trace of the last consultation
5 Explain a conclusion (HOW was it reached?)
6 Browse the knowledge base
7 Run the automated test suite
8 Help
0 Quit
```

**Figure 3:** The main menu

In forward mode the 42 questions appear in eight labelled sections. The user answers **y** or **n** (Enter alone means no), and can type **s** to skip the rest of a section. The result shows the classification, the management group, the guideline action plan and the cited trace (Figure 4).

```
Case classification : SEVERE DENGUE
Management group   : GROUP C -- requires emergency treatment and urgent referral

Because GROUP C -- requires emergency treatment and urgent referral:
* EMERGENCY. Admit to a hospital with intensive care and blood transfusion
  facilities, and refer urgently.
* Begin judicious intravenous resuscitation with isotonic crystalloid; ...

REASONING -- how this conclusion was reached
Step 2   rule r03
  IF     a pulse pressure of 20 mmHg or less
  THEN   shock
  src    [WHO2009] s.2.1.4, p.28 -- "The patient is considered to have shock if
        the pulse pressure ... is <= 20 mm Hg"
Step 6   rule r19
  IF     the dengue case definition is met
        AND severe plasma leakage
  THEN   severe dengue
  src    [WHO2009] s.2.1.4, p.27 -- "Severe dengue is defined by one or more of
        the following: (i) plasma leakage that may lead to shock ..."
```

**Figure 4:** Part of the result and explanation for scenario T04

Figure 5 shows a goal-driven session: questions arrive singly, and typing **w** at any of them reveals the goal chain behind it.

```

Goal: severe dengue
I will ask only the questions I need. Type w for why.

? Does the patient live in, or has travelled in the last 14 days to,
a dengue-endemic area?
[y = yes, n = no, w = why am I being asked this]
> w

WHY: I am working backwards through these goals.
Top goal: severe dengue
needed for the above: the dengue case definition is met
needed for the above: probable dengue
needed for the above: lives in or has travelled to a dengue-endemic area

```

**Figure 5:** A backward-chaining question with its WHY explanation

## 6. Test Cases

The suite runs 32 tests automatically (`test.bat` or menu option 7), and all of them pass. Scenario inputs are defined in `scenarios.pl`. T tests use forward chaining, B tests use backward chaining with a scenario's answers, and E1 tests the explanation facility.

| ID  | Scenario and expected result                                                                                        | Actual result                                                                                    | Pass |
|-----|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------|
| T01 | Classic dengue, fit for home care. Expected: without warning signs; Group A; rules R08, R17, R24, R32, R34, R35.    | Without warning signs; Group A; R08, R17, R24, R32, R34, R35 fired.                              | Yes  |
| T02 | Abdominal pain and persistent vomiting. Expected: with warning signs; Group B; R02, R08, R17, R23, R27, R34.        | With warning signs; Group B; R02, R08, R17, R23, R27, R34 fired.                                 | Yes  |
| T03 | No warning signs, but pregnant. Expected: without warning signs; Group B through R28, not Group A.                  | Without warning signs; Group B; R05, R08, R17, R24, R28, R34 fired; R32 did not fire.            | Yes  |
| T04 | Dengue shock syndrome. Expected: severe dengue; Group C; shock found by R03.                                        | Severe dengue; Group C; R02, R03, R08, R09, R17, R19, R22, R26, R34 fired.                       | Yes  |
| T05 | AST or ALT $\geq 1000$ IU/L. Expected: severe dengue through organ impairment (R13, R21); Group C.                  | Severe dengue; Group C; R02, R08, R13, R17, R21, R22, R26, R34 fired.                            | Yes  |
| T06 | Laboratory-confirmed, severe bleeding. Expected: case met through R07/R18; severe dengue through R11, R20; Group C. | Severe dengue; Group C; R02, R07, R11, R18, R20, R22, R26, R34 fired.                            | Yes  |
| T07 | Only one criterion present. Expected: case definition not met; no management group.                                 | Case definition not met; only R25 fired; no group assigned.                                      | Yes  |
| T08 | Laboratory-confirmed, lives alone. Expected: without warning signs; Group B through R29.                            | Without warning signs; Group B; R06, R07, R18, R24, R29, R34 fired.                              | Yes  |
| T09 | Rising haematocrit with falling platelets only. Expected: R01 gives a warning sign; Group B.                        | With warning signs; Group B; R01, R02, R08, R17, R23, R27, R34 fired.                            | Yes  |
| T10 | Recovering inpatient. Expected: all four discharge criteria met (R33).                                              | Discharge criteria met; R07, R18, R24, R32, R33, R34, R35 fired.                                 | Yes  |
| T11 | No warning signs, cannot tolerate oral fluids. Expected: Group B through R30.                                       | Without warning signs; Group B; R08, R17, R24, R30, R34 fired.                                   | Yes  |
| T12 | Defervescence on day 3–7 with rising haematocrit. Expected: critical phase detected (R16); Group B.                 | Critical phase begun; with warning signs; Group B; R01, R02, R08, R16, R17, R23, R27, R34 fired. | Yes  |
| B01 | Goal “severe dengue” with T04 answers. Expected: proved.                                                            | Proved.                                                                                          | Yes  |

| ID  | Scenario and expected result                                                                                                                                                 | Actual result                                                                    | Pass |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------|
| B02 | Goal “severe dengue” with T01 answers. Expected: not proved.                                                                                                                 | Not proved.                                                                      | Yes  |
| B03 | Goal “dengue with warning signs” with T02 answers. Expected: proved.                                                                                                         | Proved.                                                                          | Yes  |
| B04 | Goal “Group A” with T01 answers. Expected: proved.                                                                                                                           | Proved.                                                                          | Yes  |
| B05 | Goal “Group A” with T03 answers (pregnant). Expected: not proved.                                                                                                            | Not proved.                                                                      | Yes  |
| B06 | Goal “Group B” with T03 answers. Expected: proved.                                                                                                                           | Proved.                                                                          | Yes  |
| B07 | Goal “discharge ready” with T10 answers. Expected: proved.                                                                                                                   | Proved.                                                                          | Yes  |
| B08 | Goal “dengue case” with T07 answers. Expected: not proved.                                                                                                                   | Not proved.                                                                      | Yes  |
| E1  | Explanation, run on all 12 scenarios. Expected: every conclusion has a stored justification, every fired rule has a citation, and every rule in the knowledge base is cited. | 12 of 12 passed: 0 unexplained conclusions, 0 uncited rules; all 35 rules cited. | Yes  |

## 7. How to Run the System

**Code:** *DengueTriage-ES-code.zip*, submitted with this report.

**Required software:** SWI-Prolog 8.0 or later (tested on 10.0.2) from [swi-prolog.org/Download.html](http://swi-prolog.org/Download.html), or on Windows run `winget install --id SWI-Prolog.SWI-Prolog`. Nothing else has to be installed or configured, and the system runs offline.

### Steps:

1. Unzip *DengueTriage-ES-code.zip* into any folder.
2. On Windows, double-click `run.bat`; it finds SWI-Prolog even when it is not on the PATH. On macOS or Linux, run `sh run.sh`. On any platform, `swipl -g start -t halt main.pl` also works.
3. To see a prepared case, type **3**, then **t04**. Type **5** and then **1** to see the proof tree.
4. For a new patient, choose **1** to answer every question, or **2** to test one hypothesis (type **w** at any question to see why it is asked).
5. Type **0** to quit.

To run the tests, double-click `test.bat` or run `swipl -g run_tests -t halt main.pl`. Each of the three groups should end “0 failed”, followed by “every rule in the knowledge base carries a source: YES”.

**Example input (scenario T02, option 1):** answer **y** to endemic area, fever, nausea or vomiting, aches and pains, abdominal pain, persistent vomiting, tolerates oral fluids and passes urine; press Enter (no) for every other question.

**Example output:** DENGUE WITH WARNING SIGNS; GROUP B – refer for in-hospital management; the WHO fluid regimen and monitoring plan; six rules (R02, R08, R17, R23, R27, R34), each listed with its source.

## 8. References

- [1] World Health Organization, “Dengue and severe dengue,” Fact sheet, Aug. 21, 2025. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/dengue-and-severe-dengue> (accessed Sep. 2026)
- [2] National Dengue Control Unit, Ministry of Health, Sri Lanka, “Sri Lanka situation.” [Online]. Available: <https://www.dengue.health.gov.lk/sri-lanka-situation/> (accessed Sep. 2026)
- [3] World Health Organization, *Dengue: Guidelines for Diagnosis, Treatment, Prevention and Control*, new ed. Geneva, Switzerland: WHO, 2009, WHO/HTM/NTD/DEN/2009.1, ISBN 978-92-4-154787-1. [Online]. Available: <https://www.paho.org/sites/default/files/2022-08/2009-cde-dengue-guidelines-diagnosis-treatment-prevention-control.pdf>
- [4] Ministry of Health, Sri Lanka, and Ceylon College of Physicians, *National Guidelines on Management of Dengue Fever and Dengue Haemorrhagic Fever in Adults*, rev. ed. Colombo, Sri Lanka: Epidemiology Unit, Nov. 2012. [Online]. Available: <https://ccp.lk/wp-content/uploads/2024/11/Guidelines-on-Management-of-DF-DHF-in-Adults.pdf>
- [5] Centers for Disease Control and Prevention, “Clinical features of dengue” and “Guidelines for classifying dengue.” [Online]. Available: <https://www.cdc.gov/dengue/hcp/clinical-signs/index.html> and <https://www.cdc.gov/dengue/hcp/clinical-signs/guidelines.html> (accessed Sep. 2026)

## Annex A: Complete Rule Set

All 35 rules, printed from `kb.pl` by the system’s own knowledge-base browser. Every rule is taken from [3]. The source column gives the section and page; the exact sentence behind each rule is quoted in the `source/3` fact next to it in Annex B.

| Rule | IF                                                                                                                                                                                                                                            | THEN                                           | Source               |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------------|
| R01  | a rising haematocrit <b>AND</b> a rapidly falling platelet count                                                                                                                                                                              | rising haematocrit with falling platelet count | [3] Figure 1.4, p.11 |
| R02  | any one of: abdominal pain or tenderness, or persistent vomiting, or clinical fluid accumulation, or mucosal bleeding, or lethargy or restlessness, or liver enlargement greater than 2 cm, or rising haematocrit with falling platelet count | at least one warning sign is present           | [3] Figure 1.4, p.11 |
| R03  | a pulse pressure of 20 mmHg or less                                                                                                                                                                                                           | shock                                          | [3] s.2.1.4, p.28    |
| R04  | signs of poor capillary perfusion                                                                                                                                                                                                             | shock                                          | [3] s.2.1.4, p.28    |
| R05  | any one of: pregnancy, or infancy, or old age, or obesity, or diabetes mellitus, or renal failure, or chronic haemolytic disease                                                                                                              | a co-existing condition is present             | [3] s.2.3.2.2, p.34  |
| R06  | any one of: the patient lives alone, or the patient lives far from a health facility without reliable transport                                                                                                                               | social circumstances prevent safe home care    | [3] s.2.3.2.2, p.34  |
| R07  | a positive dengue laboratory test                                                                                                                                                                                                             | laboratory-confirmed dengue                    | [3] Figure 1.4, p.11 |
| R08  | lives in or has travelled to a dengue-endemic area <b>AND</b> fever <b>AND</b> at least 2 of the probable-dengue criteria                                                                                                                     | probable dengue                                | [3] Figure 1.4, p.11 |
| R09  | shock                                                                                                                                                                                                                                         | severe plasma leakage                          | [3] Figure 1.4, p.11 |
| R10  | clinical fluid accumulation <b>AND</b> respiratory distress                                                                                                                                                                                   | severe plasma leakage                          | [3] Figure 1.4, p.11 |



| Rule | IF                                                                                                                                                                                                                                                                            | THEN                                                       | Source               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|----------------------|
| R11  | severe overt bleeding with unstable haemodynamic status                                                                                                                                                                                                                       | severe bleeding                                            | [3] s.2.3.3, p.41    |
| R12  | a falling haematocrit after fluid resuscitation with continuing instability                                                                                                                                                                                                   | severe bleeding                                            | [3] s.2.3.3, p.41    |
| R13  | AST or ALT of 1000 IU/L or above                                                                                                                                                                                                                                              | severe organ impairment                                    | [3] Figure 1.4, p.11 |
| R14  | impaired consciousness                                                                                                                                                                                                                                                        | severe organ impairment                                    | [3] Figure 1.4, p.11 |
| R15  | impairment of the heart or other organs                                                                                                                                                                                                                                       | severe organ impairment                                    | [3] Figure 1.4, p.11 |
| R16  | defervescence <b>AND</b> day 3 to 7 of the illness <b>AND</b> a rising haematocrit                                                                                                                                                                                            | the critical phase has begun                               | [3] s.2.1.2, p.26    |
| R17  | probable dengue                                                                                                                                                                                                                                                               | the dengue case definition is met                          | [3] Figure 1.4, p.11 |
| R18  | laboratory-confirmed dengue                                                                                                                                                                                                                                                   | the dengue case definition is met                          | [3] Figure 1.4, p.11 |
| R19  | the dengue case definition is met <b>AND</b> severe plasma leakage                                                                                                                                                                                                            | severe dengue                                              | [3] s.2.1.4, p.27    |
| R20  | the dengue case definition is met <b>AND</b> severe bleeding                                                                                                                                                                                                                  | severe dengue                                              | [3] s.2.1.4, p.27    |
| R21  | the dengue case definition is met <b>AND</b> severe organ impairment                                                                                                                                                                                                          | severe dengue                                              | [3] s.2.1.4, p.27    |
| R22  | severe dengue                                                                                                                                                                                                                                                                 | SEVERE DENGUE                                              | [3] Figure 1.4, p.11 |
| R23  | the dengue case definition is met <b>AND</b> at least one warning sign is present <b>AND</b> NOT severe dengue                                                                                                                                                                | DENGUE WITH WARNING SIGNS                                  | [3] Figure 1.4, p.11 |
| R24  | the dengue case definition is met <b>AND</b> no warning sign <b>AND</b> NOT severe dengue                                                                                                                                                                                     | DENGUE WITHOUT WARNING SIGNS                               | [3] Figure 1.4, p.11 |
| R25  | the dengue case definition is NOT met                                                                                                                                                                                                                                         | DENGUE CASE DEFINITION NOT MET                             | [3] Figure 1.4, p.11 |
| R26  | severe dengue                                                                                                                                                                                                                                                                 | GROUP C – requires emergency treatment and urgent referral | [3] s.2.3.2.3, p.35  |
| R27  | DENGUE WITH WARNING SIGNS                                                                                                                                                                                                                                                     | GROUP B – refer for in-hospital management                 | [3] s.2.3.2.2, p.34  |
| R28  | the dengue case definition is met <b>AND</b> a co-existing condition is present <b>AND</b> NOT severe dengue                                                                                                                                                                  | GROUP B – refer for in-hospital management                 | [3] s.2.3.2.2, p.34  |
| R29  | the dengue case definition is met <b>AND</b> social circumstances prevent safe home care <b>AND</b> NOT severe dengue                                                                                                                                                         | GROUP B – refer for in-hospital management                 | [3] s.2.3.2.2, p.34  |
| R30  | the dengue case definition is met <b>AND</b> the patient CANNOT tolerate adequate oral fluids <b>AND</b> NOT severe dengue                                                                                                                                                    | GROUP B – refer for in-hospital management                 | [3] s.2.3.2.1, p.33  |
| R31  | the dengue case definition is met <b>AND</b> the patient does NOT pass urine at least once every six hours <b>AND</b> NOT severe dengue                                                                                                                                       | GROUP B – refer for in-hospital management                 | [3] s.2.3.2.1, p.33  |
| R32  | DENGUE WITHOUT WARNING SIGNS <b>AND</b> the patient tolerates adequate volumes of oral fluids <b>AND</b> the patient passes urine at least once every six hours <b>AND</b> a stable haematocrit <b>AND</b> no co-existing condition <b>AND</b> no adverse social circumstance | GROUP A – may be managed at home                           | [3] s.2.3.2.1, p.33  |
| R33  | no fever for 48 hours <b>AND</b> improvement in clinical status <b>AND</b> an increasing trend of platelet count <b>AND</b> a stable haematocrit                                                                                                                              | the patient meets the discharge criteria                   | [3] Textbox F, p.48  |

| Rule | IF                                | THEN                                | Source              |
|------|-----------------------------------|-------------------------------------|---------------------|
| R34  | the dengue case definition is met | NSAIDs and aspirin must be avoided  | [3] s.2.3.2.1, p.34 |
| R35  | GROUPE – may be managed at home   | daily ambulatory review is required | [3] s.2.3.2.1, p.34 |

## Annex B: Decision-Making Code

B.1 reproduces the rule base from `kb.pl` exactly, with each rule's citation beside it. B.2 and B.3 are the parts of `engine.pl` that put those rules to work by forward and backward chaining. Everything else is in the submitted code.

### B.1 Rule base: all 35 rules with their sources (`kb.pl`)

```
% SECTION 4 -- PRODUCTION RULES
%
%      rule(RuleId, Layer, Conclusion, PremiseList)
%
% Layer is the stratum of the rule. The forward chainer saturates
% layer N completely before it starts layer N+1. Because every
% negated premise absent(X) refers only to a strictly lower layer,
% the knowledge base is a stratified logic program and the fixpoint
% it reaches is unique and independent of rule ordering.
%
% Premise forms understood by the engine (see engine.pl):
%      Atom          -- the fact must be established
%      absent(Atom)   -- the fact must NOT be established
%      any_of(List)   -- at least one member established
%      atleast(N,Cls) -- at least N members of class Cls established
% -----
max_layer(6).

% --- Layer 1 : composite observations -----

% R01 Laboratory warning sign: rising HCT with falling platelets.
rule(r01, 1, lab_warning,
    [ hct_rising, platelet_dropping ]).
source(r01, 'WHO2009', 'Figure 1.4, p.11 -- warning signs: "Laboratory: increase in HCT concurrent with rapid decrease in platelet count"').

% R02 Any of the seven warning signs is present.
rule(r02, 1, warning_sign_present,
    [ any_of([abdominal_pain, persistent_vomiting, fluid_accumulation,
              mucosal_bleeding, lethargy_restlessness, liver_enlargement,
              lab_warning]) ]).
source(r02, 'WHO2009', 'Figure 1.4, p.11 -- the seven warning signs').

% R03 Shock recognised by a narrow pulse pressure.
rule(r03, 1, shock,
    [ narrow_pulse_pressure ]).
source(r03, 'WHO2009', 's.2.1.4, p.28 -- "The patient is considered to have shock if the pulse pressure ... is <= 20 mm Hg"').

% R04 Shock recognised by poor capillary perfusion.
rule(r04, 1, shock,
    [ poor_perfusion ]).
source(r04, 'WHO2009', 's.2.1.4, p.28 -- "or he/she has signs of poor capillary perfusion (cold extremities, delayed capillary refill, or rapid pulse rate)"').

% R05 Presence of at least one co-existing risk condition.
rule(r05, 1, coexisting_risk,
    [ any_of([pregnancy, infancy, old_age, obesity, diabetes_mellitus,
              renal_failure, chronic_haemolytic_disease]) ]).
source(r05, 'WHO2009', 's.2.3.2.2, p.34 -- "co-existing conditions that may make dengue or its management more complicated (such as pregnancy, infancy, old age, obesity, diabetes mellitus, renal failure, chronic haemolytic diseases)"').

% R06 Presence of a social circumstance that prevents safe home care.
rule(r06, 1, social_risk,
    [ any_of([lives_alone, lives_far_from_facility]) ]).
source(r06, 'WHO2009', 's.2.3.2.2, p.34 -- "certain social circumstances (such as living alone, or living far from a health facility without reliable means of transport)"').
```

```

% R07 Laboratory confirmation of dengue.
rule(r07, 1, confirmed_dengue,
    [ lab_confirmed ]).
source(r07, 'WHO2009', 'Figure 1.4, p.11 -- "Laboratory-confirmed dengue"').

% --- Layer 2 : case definition and severe criteria -----

% R08 Probable dengue.
rule(r08, 2, probable_dengue,
    [ endemic_exposure, fever, atleast(2, dengue_criterion) ]).
source(r08, 'WHO2009', 'Figure 1.4, p.11 -- "Probable dengue: live in / travel to dengue endemic area.
Fever and 2 of the following criteria: Nausea, vomiting; Rash; Aches and pains; Tourniquet test positive;
Leukopenia; Any warning sign"').

% R09 Severe plasma leakage leading to shock.
rule(r09, 2, severe_plasma_leakage,
    [ shock ]).
source(r09, 'WHO2009', 'Figure 1.4, p.11 -- "Severe plasma leakage leading to: Shock (DSS)"').

% R10 Severe plasma leakage: fluid accumulation with respiratory distress.
rule(r10, 2, severe_plasma_leakage,
    [ fluid_accumulation, respiratory_distress ]).
source(r10, 'WHO2009', 'Figure 1.4, p.11 -- "Fluid accumulation with respiratory distress"').

% R11 Severe bleeding: overt bleeding with unstable haemodynamics.
rule(r11, 2, severe_bleeding,
    [ overt_bleeding_unstable ]).
source(r11, 'WHO2009', 's.2.3.3, p.41 -- "persistent and/or severe overt bleeding in the presence of
unstable haemodynamic status, regardless of the haematocrit level"').

% R12 Severe bleeding: HCT falls after resuscitation, patient unstable.
rule(r12, 2, severe_bleeding,
    [ hct_drop_after_resuscitation ]).
source(r12, 'WHO2009', 's.2.3.3, p.41 -- "a decrease in haematocrit after fluid resuscitation together with
unstable haemodynamic status"').

% R13 Severe organ impairment: liver.
rule(r13, 2, severe_organ_impairment,
    [ ast_alt_ge_1000 ]).
source(r13, 'WHO2009', 'Figure 1.4, p.11 -- "Severe organ involvement: Liver: AST or ALT >= 1000"').

% R14 Severe organ impairment: central nervous system.
rule(r14, 2, severe_organ_impairment,
    [ impaired_consciousness ]).
source(r14, 'WHO2009', 'Figure 1.4, p.11 -- "CNS: Impaired consciousness"').

% R15 Severe organ impairment: heart and other organs.
rule(r15, 2, severe_organ_impairment,
    [ heart_impairment ]).
source(r15, 'WHO2009', 'Figure 1.4, p.11 -- "Heart and other organs"').

% R16 Onset of the critical phase.
rule(r16, 2, critical_phase_onset,
    [ defervescence, day_of_illness_3_7, hct_rising ]).
source(r16, 'WHO2009', 's.2.1.2, p.26 -- "Around the time of defervescence ... usually on days 3-7 of
illness, an increase in capillary permeability in parallel with increasing haematocrit levels may occur.
This marks the beginning of the critical phase"').

% --- Layer 3 : the dengue case -----

% R17 A probable case is a dengue case.
rule(r17, 3, dengue_case,
    [ probable_dengue ]).
source(r17, 'WHO2009', 'Figure 1.4, p.11 -- probable dengue enters the case classification').

% R18 A laboratory-confirmed case is a dengue case.
rule(r18, 3, dengue_case,
    [ confirmed_dengue ]).
source(r18, 'WHO2009', 'Figure 1.4, p.11 -- "Laboratory-confirmed dengue (important when no sign of plasma
leakage)"').

% --- Layer 4 : severity -----

% R19 Severe dengue by severe plasma leakage.
rule(r19, 4, severe_dengue,
    [ dengue_case, severe_plasma_leakage ]).
source(r19, 'WHO2009', 's.2.1.4, p.27 -- "Severe dengue is defined by one or more of the following: (i)
plasma leakage that may lead to shock (dengue shock) and/or fluid accumulation, with or without respiratory
distress"').

% R20 Severe dengue by severe bleeding.
rule(r20, 4, severe_dengue,
    [ dengue_case, severe_bleeding ]).

```

```

source(r20, 'WHO2009', 's.2.1.4, p.27 -- "and/or (ii) severe bleeding"').

% R21 Severe dengue by severe organ impairment.
rule(r21, 4, severe_dengue,
    [ dengue_case, severe_organ_impairment ] ).
source(r21, 'WHO2009', 's.2.1.4, p.27 -- "and/or (iii) severe organ impairment"').

% --- Layer 5 : case classification -----

% R22 Classification: severe dengue.
rule(r22, 5, classification(severe_dengue),
    [ severe_dengue ] ).
source(r22, 'WHO2009', 'Figure 1.4, p.11 -- third classification band "SEVERE DENGUE"').

% R23 Classification: dengue with warning signs.
rule(r23, 5, classification(dengue_with_warning_signs),
    [ dengue_case, warning_sign_present, absent(severe_dengue) ] ).
source(r23, 'WHO2009', 'Figure 1.4, p.11 -- "DENGUE with warning signs"').

% R24 Classification: dengue without warning signs.
rule(r24, 5, classification(dengue_without_warning_signs),
    [ dengue_case, absent(warning_sign_present), absent(severe_dengue) ] ).
source(r24, 'WHO2009', 'Figure 1.4, p.11 -- "DENGUE without warning signs"').

% R25 Not classifiable as dengue on the information supplied.
rule(r25, 5, classification(dengue_not_established),
    [ absent(dengue_case) ] ).
source(r25, 'WHO2009', 'Figure 1.4, p.11 -- the case definition is not met, so no dengue classification can
be assigned').

% --- Layer 6 : management group and advice -----

% R26 Group C -- emergency treatment and urgent referral.
rule(r26, 6, management_group(c),
    [ severe_dengue ] ).
source(r26, 'WHO2009', 's.2.3.2.3, p.35 -- "Group C - patients who require emergency treatment and urgent
referral when they have severe dengue"').

% R27 Group B -- warning signs present.
rule(r27, 6, management_group(b),
    [ classification(dengue_with_warning_signs) ] ).
source(r27, 'WHO2009', 's.2.3.2.2, p.34 -- "These include patients with warning signs"').

% R28 Group B -- co-existing condition.
rule(r28, 6, management_group(b),
    [ dengue_case, coexisting_risk, absent(severe_dengue) ] ).
source(r28, 'WHO2009', 's.2.3.2.2, p.34 -- "those with co-existing conditions that may make dengue or its
management more complicated"').

% R29 Group B -- social circumstances.
rule(r29, 6, management_group(b),
    [ dengue_case, social_risk, absent(severe_dengue) ] ).
source(r29, 'WHO2009', 's.2.3.2.2, p.34 -- "and those with certain social circumstances"').

% R30 Group B -- cannot tolerate oral fluids.
rule(r30, 6, management_group(b),
    [ dengue_case, absent(tolerates_oral_fluids), absent(severe_dengue) ] ).
source(r30, 'WHO2009', 's.2.3.2.1, p.33 -- Group A is restricted to "patients who are able to tolerate
adequate volumes of oral fluids", so a patient who cannot must be managed in hospital').

% R31 Group B -- inadequate urine output.
rule(r31, 6, management_group(b),
    [ dengue_case, absent(passes_urine_6h), absent(severe_dengue) ] ).
source(r31, 'WHO2009', 's.2.3.2.1, p.33 -- Group A is restricted to patients who "pass urine at least once
every six hours"').

% R32 Group A -- may be managed at home.
rule(r32, 6, management_group(a),
    [ classification(dengue_without_warning_signs),
      tolerates_oral_fluids, passes_urine_6h, stable_haematocrit,
      absent(coexisting_risk), absent(social_risk) ] ).
source(r32, 'WHO2009', 's.2.3.2.1, p.33 -- "These are patients who are able to tolerate adequate volumes of
oral fluids and pass urine at least once every six hours, and do not have any of the warning signs ...
Those with stable haematocrit can be sent home"').

% R33 Fit for discharge.
rule(r33, 6, discharge_ready,
    [ afebrile_48h, clinical_improvement, platelet_rising, stable_haematocrit ] ).
source(r33, 'WHO2009', 'Textbox F, p.48 -- "Discharge criteria (all of the following conditions must be
present): No fever for 48 hours. Improvement in clinical status ... Increasing trend of platelet count.
Stable haematocrit without intravenous fluids"').

% R34 Avoid NSAIDs in any dengue case.

```

```

rule(r34, 6, avoid_nsaid,
    [ dengue_case ]).
source(r34, 'WHO2009', 's.2.3.2.1, p.34 -- "Do not give acetylsalicylic acid (aspirin), ibuprofen or other
non-steroidal anti-inflammatory agents (NSAIDs) as these drugs may aggravate gastritis or bleeding"').

% R35 Daily review while ambulatory.
rule(r35, 6, daily_review,
    [ management_group(a) ]).
source(r35, 'WHO2009', 's.2.3.2.1, p.34 -- "Ambulatory patients should be reviewed daily for disease
progression ... until they are out of the critical period"').

```

## B.2 Premise evaluation and forward chaining (engine.pl)

```

forward_chain :- forward_chain(silent).

forward_chain(Mode) :-
    kb:max_layer(Max),
    forall(between(1, Max, Layer), saturate_layer(Layer, Mode)).

saturate_layer(Layer, Mode) :-
    ( fire_one(Layer, Mode)
    -> saturate_layer(Layer, Mode)
    ; true
    ).

fire_one(Layer, Mode) :-
    kb:rule(Id, Layer, Conclusion, Premises),
    \+ wm(Conclusion, _),
    premises_hold(Premises, Used),
    !,
    next_seq(N),
    assertz(wm(Conclusion, fired(Id, Used))),
    assertz(fired_seq(N, Id, Conclusion, Used)),
    ( Mode == verbose -> announce_firing(N, Id, Conclusion, Used) ; true ).

announce_firing(N, Id, Conclusion, Used) :-
    describe(Conclusion, CT),
    format(' [~w] ~w fires~n', [N, Id]),
    format('      premises : ~w~n', [Used]),
    format('      concludes: ~w~n', [CT]).

% --- premise evaluation -----
% premises_hold(+PremiseList, -FactsActuallyUsed)

premises_hold([], []).
premises_hold([P|Ps], Used) :-
    premise_holds(P, U1),
    premises_hold(Ps, U2),
    append(U1, U2, Used).

% A satisfied negative premise is recorded as part of the
% justification, so the trace can show that a rule fired partly
% because something was NOT established.
premise_holds(absent(X), [absent(X)]) :- !,
    \+ wm(X, _).
premise_holds(any_of(List), [F]) :- !,
    member(F, List),
    wm(F, _),
    !.
premise_holds(atleast(N, Class), Found) :- !,
    findall(F, (class_member(Class, F), wm(F, _)), Found),
    length(Found, K),
    K >= N.
premise_holds(Fact, [Fact]) :-
    wm(Fact, _).

class_member(Class, F) :-
    Goal =.. [Class, F],
    call(kb:Goal).

```

## B.3 Backward chaining (engine.pl)

```

prove(Goal, Proof) :-
    push_goal(Goal),
    catch(prove_(Goal, Proof), E, (pop_goal, throw(E))),
    pop_goal.

prove_(absent(G), node(absent(G), negation, [])) :- !,
    \+ prove_quiet(G).
prove_(any_of(List), node(any_of(List), disjunction, [Sub])) :- !,

```

```

    member(G, List),
    prove(G, Sub),
    !.
% Counting stops as soon as N members are established, so a goal
% driven session never asks for the sixth criterion once the second
% has already been confirmed.
prove_(atleast(N, Class), node(atleast(N, Class), counting, Subs)) :- !,
    findall(F, class_member(Class, F), Fs),
    collect_atleast(Fs, N, [], Subs).
prove_(Goal, node(Goal, observed, [])) :-
    wm(Goal, observed), !.
prove_(Goal, node(Goal, fired(Id), [])) :-
    wm(Goal, fired(Id, _)), !.
prove_(Goal, node(Goal, asked, [])) :-
    kb:askable(Goal, _, _), !,
    ask_user(Goal),
    wm(Goal, _).
prove_(Goal, node(Goal, rule(Id), Subs)) :-
    kb:rule(Id, _Layer, Goal, Premises),
    prove_list(Premises, Subs),
    record_derived(Goal, Id, Premises).

prove_list([], []).
prove_list([P|Ps], [S|Ss]) :-
    prove(P, S),
    prove_list(Ps, Ss).

collect_atleast(_, 0, Acc, Subs) :- !, reverse(Acc, Subs).
collect_atleast([], N, _, _) :- N > 0, !, fail.
collect_atleast([F|Fs], N, Acc, Subs) :-
    ( prove(F, S)
    -> N1 is N - 1, collect_atleast(Fs, N1, [S|Acc], Subs)
    ; collect_atleast(Fs, N, Acc, Subs)
    ).

prove_quiet(G) :- prove(G, _).

record_derived(Goal, Id, Premises) :-
    ( wm(Goal, _)
    -> true
    ; next_seq(N),
      assertz(wm(Goal, fired(Id, Premises))),
      assertz(fired_seq(N, Id, Goal, Premises))
    ).

prove_and_record(Goal, Proof) :- prove(Goal, Proof).

```